

Utkarsh Tyagi

Audio & Multimodal AI | RL & Post-training

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[LinkedIn](#) | [Google Scholar](#)

Current Research & Experience

Scale AI

Machine Learning Research Engineer

San Francisco, California

June 2025 – Present

- Lead **audio and multimodal post-training** research spanning supervised fine-tuning, reinforcement learning, reward design, and evaluation for speech-to-speech and multimodal models.
- Developed **SteerDuplex**: supervised fine-tuning and two-stage RL for steerable full-duplex speech dialogue, evaluating instruction following, vocal delivery, interruption handling, and response continuity.
- Developed **POW3R (NeurIPS 2026)**, a policy-aware rubric reward method reaching comparable performance in **2.5–4× fewer training steps**; coauthored **Rubric Dropout** and **Rubric-Guided Self-Distillation**.
- Coauthored **Audio MultiChallenge (ACL Best Resource Paper)**, **Humanity's Sixth Sense**, **Beyond Seeing**, and **SciPredict**, evaluating spoken dialogue, intuitive visual reasoning, visual tool use, and scientific prediction.
- Own technical specifications, difficulty analysis, and quality standards for multimodal data programs; build annotation tooling and containerized evaluations for GUI grounding, agentic software engineering, and visual reasoning.

Selected Publications

Full publication list: [Google Scholar](#). Paper titles link to the manuscripts.

- [Not Every Rubric Teaches Equally: Policy-Aware Rubric Rewards for RLVR](#)
NeurIPS 2026
- [SteerDuplex: Steerable Duplex Speech Dialogue Models](#)
NeurIPS 2026 Workshop RTCA | **Under submission ARR**
- [Humanity's Sixth Sense: Benchmarking Intuitive Visual Reasoning in Multimodal Models](#)
ICLR 2027 (under submission)
- [Rubric Dropout: A Simple Way to Mitigate Reward Hacking in Rubric-as-Reward RL](#)
ICLR 2027 (under submission)
- [Rubric-Guided Self-Distillation: Post-Training Without Rubric Verifiers](#)
ICLR 2027 (under submission)
- [Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction](#)
ACL 2026 | **Best Resource Paper Award**
- [Beyond Seeing: Evaluating Multimodal LLMs on Tool-Enabled Image Perception, Transformation, and Reasoning](#)
EMNLP 2026
- [SciPredict: Can LLMs Predict the Outcomes of Scientific Experiments in Natural Sciences?](#)
ICML 2026
- [Audio Hallucination Attacks: Probing the Reliability of Large Audio Language Models](#)
Interspeech 2026
- [MMAU: A Massive Multi-Task Audio Understanding and Reasoning Benchmark](#)
ICLR 2025 | **Spotlight**
- [VDGD: Mitigating LVLM Hallucinations in Cognitive Prompts by Bridging the Visual Perception Gap](#)
ICLR 2025
- [ProSE: Diffusion Priors for Speech Enhancement](#)
NAACL 2025
- [GAMA: A Large Audio-Language Model with Advanced Audio Understanding and Complex Reasoning Abilities](#)
EMNLP 2024
- [CompA: Addressing the Gap in Compositional Reasoning in Audio-Language Models](#)
ICLR 2024

Additional Industry Experience

Samsung SARC/ACL

Research Intern – Efficient GenAI Inference

San Jose, California

May 2024 – August 2024

- Optimized LLMs for resource-constrained devices, improving inference latency and memory footprint; fine-tuned models and developed retrieval-augmented generation tools over internal documentation.

Atlassian India LLP

Software Development Engineer 2

Bangalore, India

July 2021 – August 2023

- Modernized Jira's issue-create experience, achieving **99.996% frontend reliability** for over **250,000 customers** and **2 million issues daily**; reduced time to interactive from **3.9s to 1.2s**.

Atlassian India LLP

Software Engineer Intern

Bangalore, India

January 2021 – June 2021

- Reworked legacy APIs and eliminated redundant database queries, reducing modal-load latency by **95.57%** and supporting a user interface serving **1 million customers**.

Samsung R&D Institute Bangalore

Research Intern – Speech AI

Bangalore, India

May 2020 – July 2020

- Researched Bixby wake-word detection and developed an efficient, vocabulary-independent on-device keyword spotter using **LSTM-CTC**.

Academic Research Experience

GAMMA Lab, University of Maryland

Research Assistant

College Park, Maryland

August 2022 – May 2025

- Researched audio-language models, multimodal reasoning, speech enhancement, and low-resource learning with **Prof. Dinesh Manocha** and **Prof. Ramani Duraiswami**; work includes **MMAU**, **GAMA**, **CompA**, and **ProSE**, plus synthetic data augmentation for speech and NLP.

Multimodal Digital Media Analysis Lab, IIITD

Machine Learning Researcher

Delhi, India

April 2022 – March 2023

- Worked with **Dr. Rajiv Ratn Shah** on multilingual automatic speech scoring for low-resource Indian languages and context-aware implicit hate-speech detection; published at the **De-Factify workshop, AAI 2023**.

Education

University of Maryland, College Park

M.S. in Computer Science | GPA: 3.9/4.0

College Park, Maryland

August 2023 – May 2025

Delhi Technological University

B.Tech in Computer Science | GPA: 8.7/10; major GPA: 9.05/10

Delhi, India

2017 – 2021

Technical Skills

Research: Speech-to-speech; full-duplex dialogue; audio-language models; multimodal reasoning; benchmark design.

Post-training: SFT; reinforcement learning (GRPO/RLVR); rubric rewards; self-distillation; reward-hacking analysis.

Languages & Frameworks: Python, C++, JavaScript, Java, SQL; PyTorch, TensorFlow, scikit-learn, React.

Research Engineering: Evaluation harnesses; annotation tools; containerized environments; rubric grading; data quality.

Awards & Professional Service

- ACL 2026 Best Resource Paper Award** – Audio MultiChallenge (shared with coauthors).
- \$7,000 NeuroPAC Fellowship**, UMD/UMIACS – research on EEG-based brain-computer interaction using LLMs.
- Peer reviewer:** NeurIPS 2026; ICLR 2025–2026; ACL Rolling Review 2024–2026 (including recent cycles); ICASSP 2025 and 2027 (current cycle); WASPAA 2025; EMNLP 2024.
- Atlassian internal hackathon People's Choice Awards**, 2021 and 2022.

Patents

- U.S. Patent 12,505,292 B2** – scheduling abstractive summaries for multi-party communication channels; co-inventor, granted December 23, 2025.
- U.S. Application 17/936,705 (US20240111959A1)** – generating and selectively outputting abstractive summaries; co-inventor, patent pending.
- U.S. Application 18/477,176 (US20250110981A1)** – configuring and transmitting abstractive summaries to collaboration applications; co-inventor, patent pending.